

# Generation Interconnection System Impact Study Report

## For Queue Project Q14-0407

Cs Murphy Point Llc 138 kV

November 25, 2016

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0407. The Interconnection Customer proposes a 560.3 MW natural gas combined cycle facility interconnecting at the Cs Murphy Point Llc 138 kV substation in the Virginia area. The request was received on December 03, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 174 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,637,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                      | Contingency                                    | Loading (%) | Rating (MVA) | DFAX (%) |
|---------------------------------------------------------|------------------------------------------------|-------------|--------------|----------|
| Cs Murphy Pt Llc Substation-Brodhead Sw Sta 138 KV line | Heiskell-Possum Point 138 KV line              | 244.4       | 241          | 77.2     |
| Heiskell-Possum Pt 138 KV                               | Brodhead Sw Sta-Cs Murphy Point Llc 138 KV     | 194.0       | 354          | 90.0     |
| Heiskell-Cs Murphy Pt Llc 138 KV Cir. 1                 | Cs Murphy Pt Llc-Heiskell 138 KV Circuit 2     | 162.3       | 375          | 70.5     |
| Heiskell-Cs Murphy Point Llc 138 KV                     | Cs Murphy Pt Llc-Heiskell 138 KV               | 154.4       | 394          | 70.5     |
| Brodhead Sw Sta-Quantico 138 KV line                    | Heiskell-Possum Pt 138 KV                      | 131.2       | 223          | 46.7     |
| Possum Point-Brodhead Sw Sta 138 KV L                   | Heiskell-Possum Point 138 KV                   | 113.5       | 221          | 30.5     |
| Innovation-Bateville 138 KV L                           | Possum Pt 345-138 kV Autotransformer           | 111.6       | 279          | 43.0     |
| Possum Point 345-138 kV Auto                            | Bateville-Innovation 138 KV                    | 111.5       | 315          | 62.8     |
| Bateville-Novac Country Club 138 KV                     | Possum Point Substation 345-138 kV Auto        | 107.5       | 236          | 43.0     |
| Meical-Eagleville 138 KV                                | Possum Point-Heiskell Substation 138 KV        | 103.9       | 226          | 22.5     |
| 8Th St Cabe Terminal-Newdale 138 KV Circuit 2           | 8Th St Cable Terminal-Newdale 138 KV Line 1    | 103.0       | 341          | 26.4     |
| Marcus Hook Energy Lp Substation-Irvine 500 KV Line     | Guilbeau Electric-Marcus Hook Energy Lp 500 KV | 101.7       | 2,306        | 5.0      |
| Carreon-8Th Street Cable Terminal 138 KV L              | 8Th St Cable Terminal-Carreon 138 KV #2        | 101.5       | 314          | 25.7     |
| Independent Hill D.P-Eagleville 138 KV                  | Heiskell-Possum Pt 138 KV                      | 100.8       | 224          | 22.5     |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Cs Murphy Pt Llc-Brodhead Sw Sta 138 KV             | Rebuild line with higher-capacity conductor | \$4,009,000   | 91% (\$3,663,000)     |
| Heiskell-Possum Pt 138 KV                           | Rebuild line with higher-capacity conductor | \$1,273,000   | 91% (\$1,159,000)     |
| Cs Murphy Pt Llc-Heiskell 138 KV #1                 | Rebuild line with higher-capacity conductor | \$963,000     | 98% (\$948,000)       |
| Heiskell-Cs Murphy Pt Llc 138 KV                    | Rebuild line with higher-capacity conductor | \$762,000     | 98% (\$751,000)       |
| Brodhead Sw Sta-Quantico 138 KV line                | Rebuild line with higher-capacity conductor | \$421,000     | 92% (\$388,000)       |
| Brodhead Sw Sta-Possum Pt Substation 138 KV         | Rebuild line with higher-capacity conductor | \$2,644,000   | 90% (\$2,391,000)     |
| Bateville-Innovation 138 KV                         | Rebuild line with higher-capacity conductor | \$1,646,000   | 88% (\$1,445,000)     |
| Possum Point 345/138kV XFMR                         | Replace transformer with higher-rated unit  | \$9,780,000   | 89% (\$8,693,000)     |
| Novec Country Club-Bateville 138 KV line            | Reconductor line                            | \$1,328,000   | 88% (\$1,166,000)     |
| Medical-Eagleville 138 KV                           | Reconductor line                            | \$2,486,000   | 81% (\$2,022,000)     |
| 8Th St Cabe Terminal-Newdale 138 KV Circuit 2       | Reconductor line                            | \$917,000     | 90% (\$823,000)       |
| Marcus Hook Energy Lp Substation-Irvine 500 KV Line | Reconductor line                            | \$6,769,000   | 6% (\$408,000)        |
| Carreon-8Th Street Cable Terminal 138 KV L          | Reconductor line                            | \$1,440,000   | 90% (\$1,293,000)     |
| Independent Hill D.P-Eagleville 138 KV              | Reconductor line                            | \$3,059,000   | 81% (\$2,487,000)     |
|                                                     | Total allocated to project                  |               | \$27,637,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.